STAT 6611: Results by Lecture

Results are numbered as in the Summary Notes, with slide numbers in parentheses. Slides repeat across lectures, so each result is listed where it first appears. Vectors are underlined, as in the notes.

Lecture 1: Generalized inverses (slides 1–4)

- **Theorem 1.1.** For any $m \times n$ matrix A of rank $r \geq 1$, there exist nonsingular P $(m \times m)$ and Q $(n \times n)$ such that

$$PAQ = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix},$$

  where D is an $r \times r$ nonsingular diagonal matrix. The matrix $G = Q \left[\begin{smallmatrix} D^{-1} & F \\ H & B \end{smallmatrix} \right] P$ satisfies $AGA = A$ for any F, H, B of appropriate dimensions. (Proof: write $A = K_1 T_1$, with K_1 formed from r linearly independent columns of A; then $D = I_r$ works.)
- **Definition.** G is a generalized inverse of A if $AGA = A$; A^- denotes an arbitrary generalized inverse of A.
- Consequences of Theorem 1.1: every matrix has a generalized inverse; any $n \times m$ matrix is a generalized inverse of the $m \times n$ zero matrix; A has infinitely many generalized inverses unless it is nonsingular, in which case A^{-1} is the only one; P, Q, D are not unique; the construction doubles as an algorithm.

Lecture 2: Consistent equations; first $X'X$ lemmas (slides 5–8)

- **Definition.** The equations $A\underline{x} = \underline{y}$ are consistent if they have at least one solution.
- **Theorem 1.2.** G is a generalized inverse of A iff $\underline{x} = G\underline{y}$ solves $A\underline{x} = \underline{y}$ for every $\underline{y}$ for which the equations are consistent. *Remark:* the same holds for the matrix equation $AX = Y$, column by column.
- **Theorem 1.3.** All solutions to a consistent $AX = Y$ are $X = A^-Y + (I - A^-A)Z$, as Z ranges over all matrices of appropriate dimensions.
- **Lemma 1.1.** G is a generalized inverse of A iff G' is a generalized inverse of A'. **Corollary.** If A is symmetric, $(A^-)'$ is a generalized inverse of A; in particular, $[(X'X)^-]'$ is a generalized inverse of $X'X$. *Remark:* a generalized inverse of a symmetric matrix can be nonsymmetric.
- **Lemma 1.2.** $\mathrm{tr}(X'X) = 0$ implies $X = 0$.
- **Lemma 1.3.** (i) $PX'X = QX'X$ implies $PX' = QX'$; (ii) $X'XP = X'XQ$ implies $XP = XQ$.
- **Lemma 1.4.** $X'X(X'X)^-X' = X'$; that is, $X(X'X)^-$ is a generalized inverse of X'. **Corollary 1.** (i) $X'X[(X'X)^-]'X' = X'$; (ii) $X(X'X)^-X'X = X$; (iii) $X[(X'X)^-]'X'X = X$. **Corollary 2.** For any X, there exists B with $X'XB = X'$.

Lecture 3: Projections, partitioned generalized inverses, Models I–VII (slides 7–16)

- **Lemma 1.5.** If B_1 and B_2 both satisfy $X'XB = X'$, then $XB_1 = XB_2$. Corollaries:

(1) $XBX = B'X'X = X$ for any B with $X'XB = X'$;
(2) $XG_1X' = XG_2X'$ for any two generalized inverses G_1, G_2 of $X'X$;
(3) $X(X'X)^-X' = X[(X'X)^-]'X'$, i.e., $X(X'X)^-X'$ is symmetric;
(4) XB is symmetric;
(5) $P_X = XB$ is invariant to the choice of B, symmetric, equal to $X(X'X)^-X'$, satisfies $P_XX = X$, and is idempotent. P_X is called an orthogonal projection matrix.

- **Notation.** $\mathcal{C}(A)$ and $\mathcal{R}(A)$ denote the column space and row space of A. **Lemma 1.6.** (i) $\mathcal{C}(B) \subseteq \mathcal{C}(A)$ iff $B = AC$ for some C; (ii) $\mathcal{R}(B) \subseteq \mathcal{R}(A)$ iff $B = CA$ for some C.
- **Theorem 1.4.** Let $M = \left[\begin{smallmatrix} A & C \\ D & B \end{smallmatrix}\right]$.
  (i) If $\mathcal{C}(C) \subseteq \mathcal{C}(A)$ and $\mathcal{R}(D) \subseteq \mathcal{R}(A)$, then, with $Q = B - DA^-C$, a generalized inverse of M is
  $$\begin{bmatrix} A^- + A^-CQ^-DA^- & -A^-CQ^- \\ -Q^-DA^- & Q^- \end{bmatrix}.$$
  (ii) If $\mathcal{C}(D) \subseteq \mathcal{C}(B)$ and $\mathcal{R}(C) \subseteq \mathcal{R}(B)$, then, with $P = A - CB^-D$, a generalized inverse of M is
  $$\begin{bmatrix} P^- & -P^-CB^- \\ -B^-DP^- & B^- + B^-DP^-CB^- \end{bmatrix}.$$
- **Theorem 1.5.** Let $W = R + D$, where $\mathcal{C}(D) \subseteq \mathcal{C}(R)$ and $\mathcal{R}(D) \subseteq \mathcal{R}(R)$, and let $D = STU$. Then $R^- - R^-ST(T + TUR^-ST)^-TUR^-$ is a generalized inverse of W.
- **Models (§2.1).** In each, $\underline{y} = X\underline{\beta} + \underline{\epsilon}$, with X a given $n \times p$ matrix and $\underline{\beta} \in \mathbb{R}^p$.
  - Model I (least squares): no assumptions on $\underline{\epsilon}$.
  - Model II (Gauss–Markov): $\mathrm{E}(\underline{\epsilon}) = \underline{0}$, $\mathrm{Var}(\underline{\epsilon}) = \sigma^2 I$, $\sigma^2 > 0$.
  - Model III (Aitken): $\mathrm{E}(\underline{\epsilon}) = \underline{0}$, $\mathrm{Var}(\underline{\epsilon}) = \sigma^2 H$, with H a given positive definite or positive semidefinite matrix.
  - Model IV (mixed effects): $\underline{\epsilon} = Z_1\underline{b}_1 + \cdots + Z_m\underline{b}_m + \underline{e}$, with $\mathrm{Var}(\underline{b}_i) = \sigma_i^2 I$, $\mathrm{Var}(\underline{e}) = \sigma^2 I$, all mutually uncorrelated; $\sigma_1^2, \ldots, \sigma_m^2, \sigma^2$ are variance components.
  - Model V (random effects): Model IV with $X = \underline{1}$.
  - Model VI (linear variance structure): $\mathrm{Var}(\underline{\epsilon}) = \theta_1 H_1 + \cdots + \theta_m H_m$, with $H_1, \ldots, H_m$ given symmetric matrices.
  - Model VII (general mixed effects): $\mathrm{Var}(\underline{\epsilon}) = V$, whose elements are functions of $\underline{\theta}$.

  Nesting: II $\subset$ III; III, IV, V $\subset$ VI $\subset$ VII; Models II–VII are all special cases of Model I. Restricting $\underline{\beta}$ (e.g., $A\underline{\beta} = \underline{b}$ or $A\underline{\beta} \geq \underline{b}$) gives constrained versions. Notation: $p^* = \mathrm{rank}(X) \leq p$; the model is full rank when $p^* = p$.
- **Definitions (§2.2).** A linear estimator of $\underline{\lambda}'\underline{\beta}$ has the form $c + \underline{a}'\underline{y}$. Under II–VII it is unbiased if $\mathrm{E}(c + \underline{a}'\underline{y}) = \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta} \in \mathbb{R}^p$. $\underline{\lambda}'\underline{\beta}$ is estimable if some linear unbiased estimator of it exists, and nonestimable otherwise.
- **Lemma 2.1.** Under II–VII, $c + \underline{a}'\underline{y}$ is unbiased for $\underline{\lambda}'\underline{\beta}$ iff $c = 0$ and $\underline{\lambda}' = \underline{a}'X$.

Lecture 4: Estimability; least squares (slides 16–23)

- **Theorem 2.1.** Under II–VII, $\underline{\lambda}'\underline{\beta}$ is estimable iff $\underline{\lambda}' = \underline{a}'X$ for some $\underline{a}$, iff $\underline{\lambda}' \in \mathcal{R}(X)$, iff $\mathrm{rank}(X) = \mathrm{rank}\left[\begin{smallmatrix} X \\ \underline{\lambda}' \end{smallmatrix}\right]$. **Corollaries.** (1) The elements of $X\underline{\beta}$ are estimable. (2) Any linear combination of estimable functions is estimable. (3) If $p^* = p$, every $\underline{\lambda}'\underline{\beta}$ is estimable.

- **Lemma 2.2.** $\text{rank}(AB) \le \text{rank}(A)$ and $\text{rank}(AB) \le \text{rank}(B)$. **Lemma 2.3.** $\text{rank}(A'A) = \text{rank}(A)$.
- **Definition.** $\underline{\lambda}_1'\underline{\beta}, \ldots, \underline{\lambda}_r'\underline{\beta}$ are linearly independent if $\Lambda = (\underline{\lambda}_1, \ldots, \underline{\lambda}_r)$ has rank r. **Theorem 2.2.** Under II–VII, there is a set of p^* linearly independent estimable functions, and no collection of estimable functions contains more than p^* linearly independent ones.
- **Least squares (§2.3).** A least squares estimate of an estimable $\underline{\lambda}'\underline{\beta}$ is $\underline{\lambda}'\widehat{\underline{\beta}}$, where $\widehat{\underline{\beta}}$ minimizes $Q(\underline{\beta}) = (\underline{y}-X\underline{\beta})'(\underline{y}-X\underline{\beta})$. Since $\partial Q/\partial\underline{\beta} = -2X'\underline{y}+2X'X\underline{\beta}$, a minimizer must satisfy the normal equations $X'X\widehat{\underline{\beta}} = X'\underline{y}$. (Optional remark: $\partial(\underline{a}'\underline{\beta})/\partial\underline{\beta} = \underline{a}$ and $\partial(\underline{\beta}'X'X\underline{\beta})/\partial\underline{\beta} = 2X'X\underline{\beta}$.)
- **Theorem 2.3.** The normal equations are consistent.
- **Theorem 2.4.** $Q(\underline{\beta})$ attains its minimum at $\underline{\beta} = \widehat{\underline{\beta}}$ iff $\widehat{\underline{\beta}}$ solves the normal equations. (Proof: $Q(\underline{\beta}) = Q(\widehat{\underline{\beta}}) + [X(\widehat{\underline{\beta}} - \underline{\beta})]'[X(\widehat{\underline{\beta}} - \underline{\beta})]$.)
- **Theorem 2.5.** Under II–VII, $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$ for any two solutions $\widehat{\underline{\beta}}$, $\widetilde{\underline{\beta}}$ of the normal equations iff $\underline{\lambda}'\underline{\beta}$ is estimable. (Proof uses the general solution from Theorem 1.3.) **Corollary.** $X\widehat{\underline{\beta}} = X\widetilde{\underline{\beta}}$ for any two solutions.

Lecture 5: Geometry; the conjugate normal equations (slides 23–28)

- **Definitions.** $\underline{a}$ and $\underline{b}$ are orthogonal if $\underline{a}'\underline{b} = 0$; the angle θ between them satisfies $\cos\theta = \underline{a}'\underline{b}/[(\underline{a}'\underline{a})^{1/2}(\underline{b}'\underline{b})^{1/2}]$, $0 \le \theta \le \pi$. **Notation.** $\mathcal{C}^\perp(X) = \{\underline{a} : X'\underline{a} = \underline{0}\}$ (the null space of X') and $\mathcal{R}^\perp(X) = \{\underline{b} : X\underline{b} = \underline{0}\}$.
- With $\widehat{\underline{y}} = X\widehat{\underline{\beta}}$ and $\widehat{\underline{\epsilon}} = \underline{y} - \widehat{\underline{y}}$: $\underline{y} = \widehat{\underline{y}} + \widehat{\underline{\epsilon}}$ and $\widehat{\underline{y}}'\widehat{\underline{\epsilon}} = 0$.
- **Theorem 2.6.** Any $\underline{y}$ decomposes uniquely as $\underline{y} = \underline{y}_1 + \underline{y}_2$ with $\underline{y}_1 \in \mathcal{C}(X)$ and $\underline{y}_2 \in \mathcal{C}^\perp(X)$; moreover, $\underline{y}_1 = X\widehat{\underline{\beta}} = P_X\underline{y}$ and $\underline{y}_2 = \underline{y} - X\widehat{\underline{\beta}} = (I - P_X)\underline{y}$. Lecture remarks: $\widehat{\underline{y}}$ is the point of $\mathcal{C}(X)$ closest to $\underline{y}$; any $\underline{t} \in \mathcal{C}(X)$ with $X'(\underline{y} - \underline{t}) = \underline{0}$ equals $\widehat{\underline{y}}$; $I - P_X = P_Z$ for any Z whose columns span $\mathcal{C}^\perp(X)$. *Example:* $\underline{y} = (2, 3)'$ and $X = (1, 1)'$ give $\widehat{\beta} = 2.5$, $\widehat{\underline{y}} = (2.5, 2.5)'$, and $\widehat{\underline{\epsilon}} = (-0.5, 0.5)'$.
- **Corollary to Theorem 2.6.** If $\mathcal{C}(W) = \mathcal{C}(X)$, then $W\widehat{\underline{\tau}} = X\widehat{\underline{\beta}}$, where $W'W\widehat{\underline{\tau}} = W'\underline{y}$ and $X'X\widehat{\underline{\beta}} = X'\underline{y}$.
- **Theorem 2.7.** Under II–VII, $\underline{\lambda}'\underline{\beta}$ is estimable iff the conjugate normal equations $X'X\underline{r} = \underline{\lambda}$ are consistent. **Corollary.** Equivalently, iff $X'X(X'X)^-\underline{\lambda} = \underline{\lambda}$.

Lecture 6: Moments of least squares estimators; Gauss–Markov (slides 28–34)

- **Theorem 2.8.** Under II–VII, the least squares estimate of an estimable $\underline{\lambda}'\underline{\beta}$ equals $\underline{r}'X'\underline{y}$, where $\underline{r}$ is any solution to $X'X\underline{r} = \underline{\lambda}$. It is unbiased, since $\underline{a} = X\underline{r}$ gives $\underline{a}'X = \underline{r}'X'X = \underline{\lambda}'$.
- **Lemma 2.4.** $\text{cov}(L'\underline{x}, M'\underline{x}) = L'\Sigma M$, where $\Sigma = \text{Var}(\underline{x})$. **Corollary.** $\text{Var}(L'\underline{x}) = L'\Sigma L$.
- **Definition.** $\Lambda'\underline{\beta}$ is estimable if each of its elements is. If $X'XR_1 = \Lambda_1$ and $X'XR_2 = \Lambda_2$, then

$\Lambda_1'\widehat{\underline{\beta}} = R_1'X'\underline{y}$ and $\Lambda_2'\widehat{\underline{\beta}} = R_2'X'\underline{y}$, with

$$\text{Model VII:}\quad \operatorname{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = \Lambda_1'(X'X)^-X'VX(X'X)^-\Lambda_2,$$
$$\text{Model II:}\quad \operatorname{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = \sigma^2\Lambda_1'(X'X)^-\Lambda_2,$$

and the variances as the special case $\Lambda_2 = \Lambda_1$.

- **Theorem 2.9 (Gauss–Markov).** Under II–VII, if $\underline{\lambda}'\underline{\beta}$ is estimable and $\widehat{\underline{\beta}}$ is any solution to the normal equations, then $\underline{\lambda}'\widehat{\underline{\beta}}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$. Under II, its variance is uniformly less than that of any other linear unbiased estimator; that is, $\underline{\lambda}'\widehat{\underline{\beta}}$ is the BLUE. (Proof: $\operatorname{Var}(\underline{a}'\underline{y}) = \operatorname{Var}(\underline{\lambda}'\widehat{\underline{\beta}}) + \operatorname{Var}(\underline{a}'\underline{y} - \underline{\lambda}'\widehat{\underline{\beta}})$, with equality iff $\underline{a} = X\underline{r}$, so the BLUE is unique.)
- **Definitions.** A quadratic form $q(\underline{x}) = \underline{x}'A\underline{x}$ is positive definite if $q(\underline{x}) > 0$ for all $\underline{x} \neq \underline{0}$; nonnegative definite if $q(\underline{x}) \geq 0$ for all $\underline{x}$; and positive semidefinite if it is nonnegative definite but not positive definite. A symmetric matrix A takes the label of its quadratic form.
- **Theorem 2.10.** Under II, if $\underline{c} + A'\underline{y}$ is any vector of linear unbiased estimators of the elements of an estimable $\Lambda'\underline{\beta}$, then $\operatorname{Var}(\underline{c} + A'\underline{y}) - \operatorname{Var}(\Lambda'\widehat{\underline{\beta}})$ is nonnegative definite.

Lecture 7: Consequences of Gauss–Markov; estimating σ^2; constraints (slides 34–39)

- The proof of Theorem 2.10 reduces to the scalar function $\underline{\ell}'\Lambda'\underline{\beta}$, which is estimable by Corollary 2 to Theorem 2.1. **Corollary 1.** $\operatorname{tr}[\operatorname{Var}(\underline{c}+A'\underline{y})] \geq \operatorname{tr}[\operatorname{Var}(\Lambda'\widehat{\underline{\beta}})]$. **Corollary 2.** $\det[\operatorname{Var}(\underline{c}+A'\underline{y})] \geq \det[\operatorname{Var}(\Lambda'\widehat{\underline{\beta}})]$; the determinant of a covariance matrix is the generalized variance. *Optional Results 1–2:* if U, V, and $U - V$ are nonnegative definite, then $\det(U) \geq \det(U - V)$.
- **Lemma 2.5.** If $A^2 = mA$ for some scalar m, then $\operatorname{tr}(A) = m\ \operatorname{rank}(A)$. (Proof via Theorem 1.1 and $\operatorname{tr}(FG) = \operatorname{tr}(GF)$.) **Corollary.** If A is idempotent, $\operatorname{tr}(A) = \operatorname{rank}(A)$.
- **Lemma 2.6.** P_A and $I - P_A$ are symmetric and idempotent, where $P_A = A(A'A)^-A'$.
- **Lemma 2.7.** For any $n \times p$ matrix A, $\operatorname{rank}(P_A) = \operatorname{rank}(A)$ and $\operatorname{rank}(I - P_A) = n - \operatorname{rank}(A)$.
- **Lemma 2.8.** If $\mathrm{E}(\underline{x}) = \underline{\mu}$ and $\operatorname{Var}(\underline{x}) = \Sigma$, then $\mathrm{E}(\underline{x}'A\underline{x}) = \operatorname{tr}(A\Sigma) + \underline{\mu}'A\underline{\mu}$.
- Under II, $Q(\widehat{\underline{\beta}}) = \underline{y}'(I - P_X)\underline{y}$, and

$$\widehat{\sigma}^2 = (\underline{y} - X\widehat{\underline{\beta}})'(\underline{y} - X\widehat{\underline{\beta}})/(n - p^*) = \underline{y}'(\underline{y} - X\widehat{\underline{\beta}})/(n - p^*)$$

  is an unbiased estimator of σ^2.
- **Constrained models (§2.9).** $\underline{\beta}$ is restricted to $A\underline{\beta} = \underline{b}$, where A is a given $q \times p$ matrix of rank $q^* \geq 1$ and the system is assumed consistent; these are Models Ia–VIIa. An estimator is unbiased for $\underline{\lambda}'\underline{\beta}$ if its expectation is $\underline{\lambda}'\underline{\beta}$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$.
- **Lemma 2.9.** Under IIa–VIIa, $c + \underline{a}'\underline{y}$ is unbiased for $\underline{\lambda}'\underline{\beta}$ iff there exists a vector $\underline{k}$ such that $c = \underline{k}'\underline{b}$ and $\underline{\lambda}' = \underline{a}'X + \underline{k}'A$.

Lecture 8: Constrained estimability and normal equations (slides 39–44)

- **Definition.** Under IIa–VIIa, $\underline{\lambda}'\underline{\beta}$ is estimable if some $c + \underline{a}'\underline{y}$ has expectation $\underline{\lambda}'\underline{\beta}$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$. **Theorem 2.11.** Under IIa–VIIa, $\underline{\lambda}'\underline{\beta}$ is estimable iff $\underline{\lambda}' \in \mathcal{R}\begin{bmatrix} X \\ A \end{bmatrix}$.

- **Constrained least squares.** $\widehat{\underline{\beta}}$ minimizes $Q(\underline{\beta})$ over $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$. The Lagrangian $L(\underline{\beta}, \underline{\rho}) = Q(\underline{\beta}) + 2\underline{\rho}'(A\underline{\beta} - \underline{b})$ yields the constrained normal equations

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} \begin{bmatrix} \widehat{\underline{\beta}} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} X'\underline{y} \\ \underline{b} \end{bmatrix}.$$

- **Lemma 2.10.** $BW = D$ is consistent iff $\mathcal{C}(D) \subseteq \mathcal{C}(B)$, iff $\mathcal{C}^{\perp}(B) \subseteq \mathcal{C}^{\perp}(D)$, iff $\underline{a}'D = \underline{0}'$ for every $\underline{a}$ with $\underline{a}'B = \underline{0}'$.
- **Theorem 2.12.** The constrained normal equations are consistent. (Proof via Lemma 2.10.)
- **Theorem 2.13.** $Q(\underline{\beta})$ attains its minimum at $\underline{\beta} = \widehat{\underline{\beta}}$, over $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$, iff the constrained normal equations have a solution whose first component is $\widehat{\underline{\beta}}$.

Lecture 9: The bordered matrix (slides 42–47)

- **Lemma 2.11.** $\operatorname{rank}\left[\begin{smallmatrix} X'X & A' \\ A & 0 \end{smallmatrix}\right] = \operatorname{rank}\left[\begin{smallmatrix} X'X \\ A \end{smallmatrix}\right] + \operatorname{rank}(A)$. (Proof: $\mathcal{C}\left[\begin{smallmatrix} X'X \\ A \end{smallmatrix}\right]$ and $\mathcal{C}\left[\begin{smallmatrix} A' \\ 0 \end{smallmatrix}\right]$ share only the zero vector.)
- **Lemma 2.12.** (i) $\mathcal{R}\left[\begin{smallmatrix} X'X \\ A \end{smallmatrix}\right] = \mathcal{R}\left[\begin{smallmatrix} X \\ A \end{smallmatrix}\right]$; (ii) $\operatorname{rank}\left[\begin{smallmatrix} X'X \\ A \end{smallmatrix}\right] = \operatorname{rank}\left[\begin{smallmatrix} X \\ A \end{smallmatrix}\right]$. **Corollary to Lemma 2.11.** $\operatorname{rank}\left[\begin{smallmatrix} X'X & A' \\ A & 0 \end{smallmatrix}\right] = \operatorname{rank}\left[\begin{smallmatrix} X \\ A \end{smallmatrix}\right] + \operatorname{rank}(A)$.
- **Theorem 2.14.** Let $G = \left[\begin{smallmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{smallmatrix}\right]$ be any generalized inverse of $\left[\begin{smallmatrix} X'X & A' \\ A & 0 \end{smallmatrix}\right]$. Then
  - (i) $A'G_{12}'A' = A'G_{21}A' = A'$;
  - (ii) $-A'G_{22}A = -A'G_{22}'A = X'XG_{21}'A = A'G_{21}X'X = X'XG_{12}A = A'G_{12}'X'X$;
  - (iii) $AG_{11}A'$, $AG_{11}X'X$, and $X'XG_{11}A'$ are all zero matrices;
  - (iv) $X'XG_{11}X'XG_{11}X'X = X'XG_{11}X'X = X'XG_{11}'X'XG_{11}X'X = X'XG_{11}'X'X$;
  - (v) $\operatorname{tr}(X'XG_{11}) = \operatorname{rank}\left[\begin{smallmatrix} X'X \\ A \end{smallmatrix}\right] - \operatorname{rank}(A) = \operatorname{tr}(X'XG_{11}')$;
  - (vi) $X'XG_{11}X'X$ and $A'G_{22}A$ are invariant to the choice of G_{11} and G_{22}.
- **Lemma 2.13.** For any such G, $X'X = X'XG_{11}X'X - A'G_{22}A$.
- **Theorem 2.15.** Under IIa–VIIa, $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$ for the first components $\widehat{\underline{\beta}}$, $\widetilde{\underline{\beta}}$ of any two solutions to the constrained normal equations iff $\underline{\lambda}'\underline{\beta}$ is estimable. (Proof via Theorem 1.3.)

Lecture 10: Constrained conjugate normal equations; constrained Gauss–Markov (slides 48–53)

- **Corollary to Theorem 2.15.** $X\widehat{\underline{\beta}} = X\widetilde{\underline{\beta}}$ for the first components of any two solutions to the constrained normal equations.
- **Lemma 2.14.** Under IIa–VIIa, $\underline{\lambda}'\underline{\beta}$ is estimable iff $\underline{\lambda}' \in \mathcal{R}\left[\begin{smallmatrix} X'X \\ A \end{smallmatrix}\right]$.
- The constrained conjugate normal equations are

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} \begin{bmatrix} \underline{r}_1 \\ \underline{r}_2 \end{bmatrix} = \begin{bmatrix} \underline{\lambda} \\ \underline{0} \end{bmatrix}, \quad \text{that is,} \quad X'X\underline{r}_1 + A'\underline{r}_2 = \underline{\lambda} \ \text{ and } \ A\underline{r}_1 = \underline{0}.$$

Theorem 2.16. Under IIa–VIIa, $\underline{\lambda}'\underline{\beta}$ is estimable iff the constrained conjugate normal equations are consistent. (Proof via Lemma 2.10.) **Corollary.** Equivalently, iff

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}^{-} \begin{bmatrix} \underline{\lambda} \\ \underline{0} \end{bmatrix} = \begin{bmatrix} \underline{\lambda} \\ \underline{0} \end{bmatrix}.$$

- **Theorem 2.17.** Under IIa–VIIa, the least squares estimate of an estimable $\underline{\lambda}'\underline{\beta}$ is $\underline{r}_1'X'\underline{y} + \underline{r}_2'\underline{b}$, where $(\underline{r}_1, \underline{r}_2)$ is any solution to the constrained conjugate normal equations. It is unbiased: $c = \underline{r}_2'\underline{b}$ and $\underline{a} = X\underline{r}_1$ satisfy Lemma 2.9 with $\underline{k} = \underline{r}_2$.
- **Covariances under IIa.** If $X'XR_1 + A'R_2 = \Lambda_1$, $AR_1 = 0$, $X'XS_1 + A'S_2 = \Lambda_2$, and $AS_1 = 0$, then
$$\operatorname{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = \sigma^2 R_1'\Lambda_2 = \sigma^2\Lambda_1'S_1 = \sigma^2\Lambda_1'G_{11}\Lambda_2,$$
for any generalized inverse G of the bordered matrix.
- **Theorem 2.18 (constrained Gauss–Markov).** Under IIa–VIIa, if $\underline{\lambda}'\underline{\beta}$ is estimable and $\widehat{\underline{\beta}}$ is the first component of any solution to the constrained normal equations, then $\underline{\lambda}'\widehat{\underline{\beta}}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$. Under IIa, it is the BLUE.
- **Estimating σ^2 under IIa.** For any solution $(\widehat{\underline{\beta}}, \underline{\rho})$ of the constrained normal equations, $Q(\widehat{\underline{\beta}}) = \underline{y}'\underline{y} - \widehat{\underline{\beta}}'X'\underline{y} - \underline{\rho}'\underline{b}$, and $\mathrm{E}[Q(\widehat{\underline{\beta}})] = \sigma^2\{n - \operatorname{rank}\left[\begin{smallmatrix} X \\ A \end{smallmatrix}\right] + \operatorname{rank}(A)\}$. Hence $\widehat{\sigma}^2 = Q(\widehat{\underline{\beta}})/(n-f)$ is unbiased, where $f = \operatorname{rank}\left[\begin{smallmatrix} X \\ A \end{smallmatrix}\right] - \operatorname{rank}(A)$.
- **Inequality constraints.** $A\underline{\beta} \geq \underline{b}$ is handled with surplus variables: $A\underline{\beta} = \underline{b} + \underline{u}$ with $u_i \geq 0$. The problem reduces to minimizing
$$\big[\underline{y} - XG_{11}X'\underline{y} - XG_{12}(\underline{b} + \underline{u})\big]'\big[\underline{y} - XG_{11}X'\underline{y} - XG_{12}(\underline{b} + \underline{u})\big]$$
over $\underline{u}$, subject to $(I - AA^-)(\underline{b} + \underline{u}) = \underline{0}$ and $u_i \geq 0$: a quadratic programming problem.

Lecture 11: Model III (slides 54–60)

- **Lemma 2.15.** If A is $n \times n$ nonnegative definite of rank $r \geq 1$, then $A = S'S$ for some $r \times n$ matrix S, and any such S has rank r. (Proof via the spectral decomposition.)
- **Transformed model** (H nonsingular). Write $H = R'R$ with R nonsingular, and let $\underline{z} = (R^{-1})'\underline{y}$, $W = (R^{-1})'X$, and $\underline{h} = (R^{-1})'\underline{\epsilon}$. Then $\underline{z} = W\underline{\beta} + \underline{h}$ with $\mathrm{E}(\underline{h}) = \underline{0}$ and $\operatorname{Var}(\underline{h}) = \sigma^2 I$, the form of Model II. Since $\mathcal{R}(W) = \mathcal{R}(X)$, the transformed model and Model III have the same estimable functions and the same linear unbiased estimators.
- **Aitken equations.** $X'H^{-1}X\widehat{\underline{\beta}} = X'H^{-1}\underline{y}$, equivalently $W'W\widehat{\underline{\beta}} = W'\underline{z}$. **Theorem 2.19.** Under III, if $\underline{\lambda}'\underline{\beta}$ is estimable and $\widehat{\underline{\beta}}$ is any solution to the Aitken equations, then $\underline{\lambda}'\widehat{\underline{\beta}}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, with variance uniformly less than that of any other linear unbiased estimator. $\widehat{\underline{\beta}}$ solves the Aitken equations iff it minimizes $(\underline{y} - X\underline{\beta})'H^{-1}(\underline{y} - X\underline{\beta})$; minimizing $(\underline{y} - X\underline{\beta})'A(\underline{y} - X\underline{\beta})$ for a positive definite A is called generalized least squares. Also,
$$\operatorname{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = \sigma^2\Lambda_1'(X'H^{-1}X)^-\Lambda_2, \qquad \widehat{\sigma}^2 = (\underline{y} - X\widehat{\underline{\beta}})'H^{-1}(\underline{y} - X\widehat{\underline{\beta}})/(n - p^*).$$
- **Possibly singular H.** For a linear estimator, $\operatorname{Var}(c + \underline{a}'\underline{y}) = \sigma^2\underline{a}'H\underline{a}$, so a BLUE minimizes $\underline{a}'H\underline{a}$ subject to $X'\underline{a} = \underline{\lambda}$. The Lagrangian $\underline{a}'H\underline{a} + 2\underline{\rho}'(X'\underline{a} - \underline{\lambda})$ gives the equations
$$\begin{bmatrix} H & X \\ X' & 0 \end{bmatrix}\begin{bmatrix} \underline{a} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} \underline{0} \\ \underline{\lambda} \end{bmatrix}, \tag{1}$$
which, for estimable $\underline{\lambda}'\underline{\beta}$, are consistent by Theorem 2.12 (the same structure as the constrained normal equations, with $H = R'R$ in place of $X'X$ and X' in place of A).
- **Theorem 2.20.** Under III, $c + \underline{a}'\underline{y}$ is a BLUE of an estimable $\underline{\lambda}'\underline{\beta}$ iff $c = 0$ and $\underline{a}$ is the first component of some solution to (1).

Lecture 12: More Model III; when the SLSE is a BLUE (slides 61–66)

- **Corollary 1 (to Theorem 2.20).** Under III, $c + \underline{a}'\underline{y}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ iff $H\underline{a} \in \mathcal{C}(X)$ and $\mathrm{E}(c + \underline{a}'\underline{y}) = \underline{\lambda}'\underline{\beta}$. *Remark:* $\underline{a}'\underline{y}$ is a BLUE of its expectation iff $H\underline{a} \in \mathcal{C}(X)$. **Corollary 2.** If $c+\underline{a}'\underline{y}$ and $k+\underline{b}'\underline{y}$ are both BLUEs of the same estimable function, then $\mathrm{Var}[c+\underline{a}'\underline{y}-(k+\underline{b}'\underline{y})] = 0$; equivalently, they are equal with probability 1.
- Let $G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$ be any generalized inverse of $\begin{bmatrix} H & X \\ X' & 0 \end{bmatrix}$. For estimable $\underline{\lambda}'\underline{\beta}$, both $\underline{\lambda}'G_{12}'\underline{y}$ and $\underline{\lambda}'G_{21}\underline{y}$ are BLUEs (equal with probability 1). For estimable $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$, the elements of $\Lambda_1'G_{12}'\underline{y}$ and $\Lambda_2'G_{12}'\underline{y}$ are BLUEs, with

$$\mathrm{cov}[\Lambda_1'G_{12}'\underline{y}, \Lambda_2'G_{12}'\underline{y}] = \sigma^2\Lambda_1'(-G_{22})\Lambda_2, \qquad \widehat{\sigma}^2 = \underline{y}'G_{11}\underline{y}/[\mathrm{rank}(H, X) - \mathrm{rank}(X)].$$

- **Model IIIa as Model III.** Treat $\underline{b}$ as q additional data points:

$$\begin{bmatrix} \underline{y} \\ \underline{b} \end{bmatrix} = \begin{bmatrix} X \\ A \end{bmatrix} \underline{\beta} + \begin{bmatrix} \underline{\epsilon} \\ \underline{0} \end{bmatrix}, \qquad \mathrm{Var}\begin{bmatrix} \underline{\epsilon} \\ \underline{0} \end{bmatrix} = \sigma^2 \begin{bmatrix} H & 0 \\ 0 & 0 \end{bmatrix}.$$

  BLUEs computed under this unconstrained Model III are BLUEs under the original constrained model, and vice versa.
- **SLSE vs. BLUE (§2.11).** The simple least squares estimator (SLSE) of an estimable $\underline{\lambda}'\underline{\beta}$ is $\underline{\lambda}'\widetilde{\underline{\beta}}$, where $X'X\widetilde{\underline{\beta}} = X'\underline{y}$. It is unbiased but in general not a BLUE. **Theorem 2.21.** Under III, the SLSE of an estimable $\underline{\lambda}'\underline{\beta}$ is a BLUE iff there exists a vector $\underline{q}$ such that $HX\underline{r} = X\underline{q}$, where $\underline{r}$ is any solution to $X'X\underline{r} = \underline{\lambda}$. **Corollary 1.** The SLSE is a BLUE for every estimable $\underline{\lambda}'\underline{\beta}$ iff there exists a matrix Q such that $HX = XQ$. **Corollary 2.** Equivalently, iff $HP_X = P_XH$ (i.e., HP_X is symmetric). *Remarks:* these results still hold with $\mathrm{Var}(\underline{y}) = \sigma^2H$ in place of H, and under Model VII with V in place of H (i.e., $VX\underline{r} = X\underline{q}$; $VX = XQ$; $VP_X = P_XV$), where the elements of $\underline{q}$ and Q may be functions of $\underline{\theta}$. See Zyskind (1967), *Annals of Mathematical Statistics* 38, 1092–1109.

Lecture 13: An example; best linear unbiased prediction (slides 65–72)

- **Example (several variates per individual).** Observe s variates on each of r individuals, with $\underline{y}_i = U\underline{\beta}_i + \underline{\epsilon}_i$ for variate i (U a given $r \times t$ matrix). Observations on the same individual may be correlated; observations on different individuals are uncorrelated. Stacking gives Model VII with $X = \mathrm{diag}(U, \ldots, U)$ and V having (i, i') block $\sigma_{ii'}I_r$. Then $VX = XQ$, where Q has (i, i') block $\sigma_{ii'}I_t$, so least squares fitted separately to each variate gives BLUEs.
- **BLUP (§2.12).** Under Model III with H nonsingular, predict $\underline{\lambda}'\underline{\beta}+u$, where $\mathrm{E}(u) = 0$, $\mathrm{Var}(u) < \infty$, and $\mathrm{cov}(\underline{\epsilon}, u) = \sigma^2\underline{k}$. A linear predictor $c + \underline{a}'\underline{y}$ is unbiased if $\mathrm{E}(c + \underline{a}'\underline{y}) = \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta}$; $\underline{\lambda}'\underline{\beta}+u$ is predictable if such a predictor exists. With $\widehat{\underline{\beta}}$ any solution to the Aitken equations and $\widehat{u} = \underline{k}'H^{-1}(\underline{y} - X\widehat{\underline{\beta}})$,

$$\underline{\lambda}'\widehat{\underline{\beta}} + \widehat{u} = \underline{b}'\underline{y}, \qquad \underline{b}' = \underline{k}'H^{-1} + (\underline{\lambda}' - \underline{k}'H^{-1}X)(X'H^{-1}X)^-X'H^{-1},$$

  is the unique best linear unbiased predictor: it has uniformly smaller mean squared error than any other linear unbiased predictor. Motivation: under joint normality, $\mathrm{E}(u \mid \underline{\epsilon}) = \underline{k}'H^{-1}\underline{\epsilon}$.

Lecture 14: Multivariate normal distributions (slides 73–80)

- **Definition.** The moment generating function (mgf) of $\underline{x}$ is $M_{\underline{x}}(\underline{t}) = \mathrm{E}[\exp(\underline{t}'\underline{x})]$, provided it exists for $-h < t_i < h$ for some $h > 0$. **Theorem 3.1.** Two random vectors whose mgfs exist have the same cdf iff their mgfs agree on an open rectangle containing the origin. **Theorem 3.2.** $\underline{x}_1, \ldots, \underline{x}_m$ are mutually independent iff $M_{\underline{x}}(\underline{t}) = \prod_{i=1}^m M_{\underline{x}_i}(\underline{t}_i)$ on an open rectangle containing the origin.
- **Theorem 3.3.** If A is symmetric and B is positive definite, then

$$\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} (\underline{x}'A\underline{x} + \underline{x}'\underline{a} + a_0) \exp\bigl[-(\underline{x}'B\underline{x} + \underline{x}'\underline{b} + b_0)\bigr]\, dx_1 \cdots dx_n$$
$$= \tfrac{1}{2}\pi^{n/2}|B|^{-1/2}\Bigl[\mathrm{tr}(AB^{-1}) - \underline{b}'B^{-1}\underline{a} + \tfrac{1}{2}\underline{b}'B^{-1}AB^{-1}\underline{b} + 2a_0\Bigr] \exp\Bigl(\tfrac{1}{4}\underline{b}'B^{-1}\underline{b} - b_0\Bigr).$$

- **Standard normal.** A scalar z has pdf $(2\pi)^{-1/2}e^{-z^2/2}$ and mgf $e^{t^2/2}$; a k-variate standard normal $\underline{z}$ has pdf $(2\pi)^{-k/2}\exp(-\frac{1}{2}\underline{z}'\underline{z})$ and mgf $\exp(\frac{1}{2}\underline{t}'\underline{t})$.
- **Definition.** Let $\mathrm{E}(\underline{x}) = \underline{\mu}$ and $\mathrm{Var}(\underline{x}) = \Sigma$ with $\mathrm{rank}(\Sigma) = k \geq 1$, and let Γ be $k \times p$ with $\Sigma = \Gamma'\Gamma$. Then $\underline{x}$ has a p-variate normal distribution of rank k, written $\underline{x} \sim N(\underline{\mu}, \Sigma)$, if its cdf is that of $\underline{\mu} + \Gamma'\underline{z}$, where $\underline{z}$ is k-variate standard normal. Its mgf is $\exp(\underline{t}'\underline{\mu} + \frac{1}{2}\underline{t}'\Sigma\underline{t})$. *Remarks:* Lemma 2.15 supplies Γ; any A with $A'A = \Sigma$ works in its place; $\underline{x}$ has no pdf on $\mathbb{R}^p$ unless $\mathrm{rank}(\Sigma) = p$.
- **Theorem 3.4.** If $\mathrm{rank}(\Sigma) = p$, then $\underline{x}$ has pdf $(2\pi)^{-p/2}|\Sigma|^{-1/2}\exp\bigl\{-\frac{1}{2}(\underline{x} - \underline{\mu})'\Sigma^{-1}(\underline{x} - \underline{\mu})\bigr\}$.
- **Theorem 3.5.** $B\underline{x} + \underline{b} \sim N(B\underline{\mu} + \underline{b},\, B\Sigma B')$. *Remark:* with Γ a $k \times p$ matrix of rank k and $\Sigma = \Gamma'\Gamma$, $(\Gamma\Gamma')^{-1}\Gamma(\underline{x} - \underline{\mu})$ is k-variate standard normal.
- **Theorem 3.6.** For conformal partitions, $\underline{x}_i \sim N(\underline{\mu}_i, \Sigma_{ii})$.
- **Theorem 3.7.** $\underline{x}_1, \ldots, \underline{x}_m$ are mutually independent iff $\Sigma_{ij} = 0$ for all $i \neq j$. (Proof: the mgf factors.)

Lecture 15: Conditionals; quadratic forms; noncentral χ^2 (slides 79–88)

- **Corollary to Theorem 3.7.** $B_1\underline{x} + \underline{b}_1$ and $B_2\underline{x} + \underline{b}_2$ are independent iff $B_1\Sigma B_2' = 0$.
- **Theorem 3.8.** For $M = \left[\begin{smallmatrix} A & C \\ D & B \end{smallmatrix}\right]$ with M, A, B square: $|M| = |B|\,|A - CB^{-1}D|$ if B is nonsingular, and $|M| = |A|\,|B - DA^{-1}C|$ if A is nonsingular.
- **Theorem 3.9.** If Σ is nonsingular, then

$$\underline{x}_1 \mid \underline{x}_2 \sim N\bigl(\underline{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\underline{x}_2 - \underline{\mu}_2),\ \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\bigr).$$

- **Example 1.** If $\underline{z} \sim N(\underline{0}, I_p)$, then $\underline{z} - \bar{z}\underline{1} = (I - \underline{1}\underline{1}'/p)\underline{z} \sim N(\underline{0},\, I - \underline{1}\underline{1}'/p)$, a p-variate normal of rank $p - 1$ (rank equals trace for an idempotent matrix).
- **Example 2.** For $\underline{x} \sim N(\underline{\mu}, \Sigma)$ with $\underline{\mu} = (1, -1, 2, 0)'$ and $\Sigma = \left[\begin{smallmatrix} 2 & 0 & 1 & 0 \\ 0 & 3 & 0 & 2 \\ 1 & 0 & 5 & 0 \\ 0 & 2 & 0 & 3 \end{smallmatrix}\right]$: $x_1 + 2x_2 + 3x_3 + x_4 + 2 \sim N(7, 76)$; $(x_1, x_3)' \sim N\bigl((1, 2)', \left[\begin{smallmatrix} 2 & 1 \\ 1 & 5 \end{smallmatrix}\right]\bigr)$; and $(x_1, x_2)' \mid (x_3, x_4)'$ is normal with mean $\bigl(\frac{1}{5}(x_3 + 3),\ \frac{2}{3}x_4 - 1\bigr)'$ and covariance $\mathrm{diag}(9/5, 5/3)$, which is smaller than the marginal covariance $\mathrm{diag}(2, 3)$ by the positive definite $\mathrm{diag}(1/5, 4/3)$.

- **Quadratic forms (§3.2).** For $\underline{x} \sim N(\underline{\mu}, \Sigma)$ and symmetric A, for $|t|$ sufficiently small,

$$M_{\underline{x}'A\underline{x}}(t) = |I - 2tA\Sigma|^{-1/2} \exp\big[t\underline{\mu}'(I - 2tA\Sigma)^{-1}A\underline{\mu}\big],$$
$$\mathrm{Var}(\underline{x}'A\underline{x}) = 2\,\mathrm{tr}\big[(A\Sigma)^2\big] + 4\underline{\mu}'A\Sigma A\underline{\mu}.$$

  Lemma 3.1. If A is positive definite and B is symmetric, there exists $\delta > 0$ such that $A + tB$ is positive definite for $|t| < \delta$. **Theorem 3.10.** If R and T are nonsingular, $|R + STU| = |R|\,|T|\,|T^{-1} + UR^{-1}S|$. **Corollary.** $|I + SU| = |I + US|$.
- **Noncentral χ^2 (§3.3).** $u \sim \chi^2(n, \lambda)$, with $\lambda \geq 0$ and $n > 0$, has pdf

$$\chi^2(u : n, \lambda) = \sum_{j=0}^{\infty} \frac{e^{-\lambda}\lambda^j}{j!} \frac{u^{(n+2j-2)/2}e^{-u/2}}{\Gamma\left(\frac{n+2j}{2}\right) 2^{j+n/2}} = \sum_{j=0}^{\infty} \frac{e^{-\lambda}\lambda^j}{j!}\, \chi^2(u : n + 2j), \qquad u > 0,$$

  and 0 for $u \leq 0$: a Poisson(λ) mixture of central $\chi^2(n + 2j)$ pdfs. Taking $\lambda = 0$ gives the central $\chi^2(n)$. The mgf is $M_u(t) = (1 - 2t)^{-n/2} \exp\{2t\lambda/(1 - 2t)\}$ for $t < \frac{1}{2}$. **Theorem 3.11.** If $\underline{x} \sim N(\underline{\mu}, I_n)$, then $\underline{x}'\underline{x} \sim \chi^2(n, \frac{1}{2}\underline{\mu}'\underline{\mu})$. Consequently, $\mathrm{E}(u) = n + 2\lambda$ and $\mathrm{Var}(u) = 2(n + 4\lambda)$. **Theorem 3.12.** If $u_1, \ldots, u_m$ are mutually independent with $u_i \sim \chi^2(n_i, \lambda_i)$, then $u_1 + \cdots + u_m \sim \chi^2(n_1 + \cdots + n_m,\ \lambda_1 + \cdots + \lambda_m)$.

In the Summary Notes, beyond Lecture 15

- **Theorem 3.13.** If $\underline{x} \sim N(\underline{\mu}, \Sigma)$ with $\underline{x}$ $n \times 1$ and $\mathrm{rank}(\Sigma) = n$, then $\underline{x}'\Sigma^{-1}\underline{x} \sim \chi^2(n, \frac{1}{2}\underline{\mu}'\Sigma^{-1}\underline{\mu})$.
- **Definition.** If $u_1 \sim \chi^2(n_1, \lambda)$ and $u_2 \sim \chi^2(n_2)$ are independent, then $w = (u_1/n_1)/(u_2/n_2)$ has a noncentral F distribution with n_1 and n_2 degrees of freedom and noncentrality parameter λ; $\lambda = 0$ gives the central F. Neither has an mgf.
- Quadratic forms $\underline{x}'A\underline{x}$ below are taken with A symmetric. **Theorem 3.14.** If $\underline{x} \sim N(\underline{\mu}, \Sigma)$, a sufficient condition for $\underline{x}'A\underline{x}$ and $B\underline{x}$ to be independent is $B\Sigma A = 0$. **Theorem 3.15.** A sufficient condition for $\underline{x}'A_1\underline{x}, \ldots, \underline{x}'A_k\underline{x}$ to be mutually independent is $A_i\Sigma A_j = 0$ for all $i \neq j$. **Corollary.** A sufficient condition for $\underline{x}'A\underline{x}$ and $\underline{x}'B\underline{x}$ to be independent is $B\Sigma A = 0$. *Remarks:* if Σ is nonsingular, the condition $B\Sigma A = 0$ in Theorem 3.14 and the Corollary is necessary as well as sufficient; if Σ is singular, it is not necessary.